THE EMPEROR JUSTINIAN IN LOD

THE PROJECT

This ontology-based project was developed within the course *Information Science and Cultural Heritage* at the University of Bologna, taught by Professor Marilena Daquino and Professor Francesca Tomasi. The project focuses on the historical figure of **Emperor Justinian I** (527–565 CE), one of the most significant rulers of the Byzantine Empire. His reign marked a crucial moment in the history of Late Antiquity, characterized by ambitious political, legal, military, and architectural initiatives that profoundly shaped the development of the Mediterranean world and influenced later European legal and cultural traditions.

The aim of this project is to explore and represent the historical and cultural legacy of Justinian through the creation of a structured semantic dataset based on Linked Open Data methodologies. By collecting and organizing cultural heritage resources related to the emperor and his historical context, the project seeks to model the relationships between historical figures, places, events, and cultural productions connected to his reign.

Linked Open Data constitutes one of the foundations of the Semantic Web, an evolution of the traditional web that aims to make information not only accessible to humans but also understandable by machines. Within this framework, the metadata describing the selected cultural heritage resources are transformed into semantic triples composed of a subject, a predicate, and an object. This process, based on RDF technologies, allows the creation of a network of interconnected data that can be explored, reused, and integrated with other datasets, enhancing both the accessibility and the interpretative potential of cultural heritage information.

STUDY OF THE DOMAIN

IDEA

The project stems from a collective reflection on the role of historical figures and monuments that function as points of intersection between different cultural traditions and historical narratives. Among the many figures of Late Antiquity, **Emperor Justinian I** emerged as a particularly compelling subject of study. More than many other rulers of the sixth century, Justinian embodies the encounter and tension between **East and West**, between the Roman imperial heritage and the evolving Byzantine world.

His reign (527–565 CE) represents one of the most ambitious attempts to restore the authority and unity of the Roman Empire. Through major military campaigns aimed at reconquering former Western territories, extensive legal reforms culminating in the *Corpus Iuris Civilis*, and monumental architectural enterprises such as the construction of **Hagia Sophia**, Justinian left a profound mark on the political, legal, and cultural landscape of the Mediterranean world. These initiatives produced a wide range of material and textual traces (e.g. laws, chronicles, monuments, artworks) that still shape historical memory today.

The resonance of Justinian's legacy extended far beyond his lifetime. His figure was remembered and reinterpreted across centuries, becoming part of both historical scholarship and literary tradition. A notable example is found in **Dante Alighieri's *Paradiso***, where Justinian himself appears as a speaking voice recounting the history and legitimacy of the Roman Empire. In *Paradiso* (Canto VI), Dante places the emperor among the spirits of the heaven of Mercury, presenting him as the ruler who restored the Roman laws and unified them under a coherent legal system. This literary reference illustrates how Justinian's political and juridical achievements continued to shape medieval intellectual thought.

From medieval literature to modern historiography, Justinian's image has been continuously reinterpreted. Contemporary media, ranging from cinema and music to games and digital culture, continue to revisit the emperor and the world he ruled, demonstrating how a sixth-century sovereign still informs discussions about law, empire, religion, and the cultural exchange between East and West.

ITEMS

In order to ground the project in tangible cultural heritage resources, a selection of **ten items** was identified. These items represent different types of cultural objects and sources, including artworks, architectural monuments, manuscripts, historical texts, audiovisual media, and digital cultural productions. The goal of this selection is to illustrate the variety of materials through which Justinian's historical legacy has been preserved, transmitted, and reinterpreted over time.

The corpus includes material evidence directly connected to Justinian's reign, such as artistic representations, architectural monuments, and legal or historiographical texts produced during or shortly after his rule. At the same time, the dataset also incorporates later cultural reinterpretations of the emperor and his historical context, demonstrating the long-lasting influence of his image in modern cultural production.

The items were selected from various digital collections and repositories provided by cultural institutions and online platforms, including museum collections, digital libraries, architectural databases, and media archives. This diversity of sources allows the project to explore Justinian's legacy from multiple perspectives, historical, artistic, legal, and cultural, while also providing a heterogeneous dataset suitable for semantic modeling and Linked Open Data representation.

KNOWLEDGE ORGANIZATION

METADATA ANALYSIS

The first phase of the project consisted in identifying and analyzing the **metadata standards** adopted by the institutions providing the selected items. Since the corpus includes materials originating from different cultural heritage domains, such as **museum collections**, **architectural databases**, **digital libraries**, **audiovisual platforms**, and **online archives**, the metadata structures used to describe these resources vary considerably. This heterogeneity required a careful examination of how each institution organizes and presents **descriptive information**.

Whenever possible, the project relied on the **metadata standards explicitly declared** by the data providers. For instance, **digital libraries** and **archival platforms** typically structure their resources following widely used **bibliographic standards**, often inspired by **Dublin Core** or similar frameworks for the description of textual and documentary resources. These standards provide essential descriptive fields such as **title**, **creator**, **date**, **subject**, **language**, and **format**, ensuring **interoperability** between collections and facilitating the integration of data into a **semantic environment**.

In the case of **museum collections** and **cultural heritage objects**, metadata reflect **domain-specific descriptive practices**. For cultural heritage objects from Italian sources, the project adopted the **ArCo ontology network**, developed by the Italian Ministry of Cultural Heritage for the structured description of national cultural resources. For artworks and cultural objects more generally, the **CIDOC CRM ontology schema** was used as the primary conceptual reference model. This ontology provides a formal framework for representing **cultural events**, **objects**, **actors**, and their relationships, including detailed information about **production**, **materials**, **techniques**, **measurements**, and **provenance**, enabling a rich semantic representation of cultural entities.

A particular case within cultural heritage objects is represented by **numismatic objects**, such as the gold solidus from the British Museum. Due to their specific descriptive requirements, such as **denomination**, **mint**, and **issuing authority**, the project adopted the **Nomisma Ontology (NMO)**, a domain-specific vocabulary designed for the semantic description of coins and monetary objects.

For **architectural heritage resources**, metadata emphasize **spatial**, **historical**, and **architectural information**. Databases dedicated to architectural documentation typically include structured descriptions concerning **construction periods**, **locations**, **architectural typologies**, **materials**, and **historical transformations**. These elements are essential for representing buildings not only as physical structures but also as **cultural and historical entities** embedded within a broader landscape.

A further challenge emerged in the case of **digital media platforms** and **contemporary cultural productions** (e.g., IMDb, YouTube, BoardGameGeek, Civilization Fandom Wiki), where no official metadata standard is declared and the available data is often limited to basic fields such as **title**, **release date**, **creator**, and **genre**. In these cases, the project mapped the available fields to established standards, primarily **Schema.org** and **Dublin Core**, selecting properties that best captured the **semantic meaning** of the original data.

When a metadata standard was not explicitly declared, the project adopted the standard most commonly associated with that type of cultural resource. This decision was guided by the need to maintain **semantic coherence** across the dataset while ensuring that each item could be accurately described within a recognized **metadata framework**.

This preliminary analysis of metadata structures played a crucial role in the project, as it provided the foundation for the subsequent phases of **theoretical modeling** and **conceptual modeling**, where the descriptive elements identified in the original metadata were

formalized and mapped to **classes** and **properties** from existing **ontologies** and **Semantic Web vocabularies**.

THEORETICAL MODEL

The **theoretical model** was built starting from the **natural language descriptions** of each **cultural heritage item**. They were collected from the institutions' metadata and organized into **subject–predicate–object statements** in **CSV format**, allowing it to be human-readable. This structure made it possible to clearly represent the connections between the items and the central figure of **Emperor Justinian I**, as well as the relationships among the items themselves. It is worth mentioning that this process gave the team a better view and understanding of what the scope of the project is. In the later stages, when the maps were created using **Draw.io** as a tool, it gave a broader perspective on possible errors or shortcomings within the data itself.

Not all data sources provided **explicit metadata standards**. In cases where an institution did not declare a formal schema, the predicates were adapted to ensure readability and to facilitate the recognition of **controlled authority forms**. Where institutions did provide structured metadata, those original fields were preserved or mapped to equivalent **semantic relationships**.

Additional triples were added to enrich the dataset, linking items to relevant **historical figures** (e.g., Theodora, Belisarius), **places** (Constantinople, Ravenna), and **concepts** (Byzantine Empire, Corpus Iuris Civilis). These extra connections were included to make the knowledge graph as interconnected as possible and to illustrate the broader cultural and historical network surrounding Justinian.

Since the creation of the diagrams manually was a demanding task, the team experimented with different diagram-making tools to see which one can automate the process for us. Having chosen **Draw.io** at last, we used the triplets in our CSV files to create

triplets in a format which was **machine readable**. That allowed us to create the diagrams in a matter of seconds, and gave us the possibility to quickly discard and edit them if needed. The triplets were used in the following format of "**X -> Y -> Z**", and ultimately used as text input in Draw.io.

The **design of the maps** was of importance to us. First on a **visual clarity** level, and secondly, on a purely **aesthetic** basis. To support visual clarity in the final diagram, entities were categorized using **distinct shapes and colors**. For example, **items** (the main cultural objects) were distinguished from **external entities**, and **literal values** (dates, measurements, titles) were visually separated from **URI-based resources**. Aesthetic-wise, the colors were picked from the pallet used in the **mosaics in Ravenna**, in order to give a sense of familiarity to the audience.

CONCEPTUAL MODEL

The **conceptual model** formalizes the relationships identified in the theoretical model by assigning each **entity**, **property**, and **class** to appropriate **Semantic Web vocabularies and ontologies**. The goal is to make the dataset **interoperable** and **machine readable** while maintaining semantic coherence across diverse cultural heritage domains.

All **project-specific entities** (such as Justinian I, Theodora, Belisarius, Corpus Iuris Civilis, and the items themselves) were assigned **custom URIs** under the project namespace sits:, defined as: **https://github.com/Justinian-LOD/Justinian-LOD/** (placeholder – adjust to actual project URI).

For each predicate expressed in natural language, the most suitable **property** was selected from **existing ontologies** based on the nature of the relationship and the type of object involved. The choice of ontology was guided by the domain of the resource being described:

- **Dublin Core** (dcterms) – for generic descriptive metadata such as title, date, language, source, and format.
- **CIDOC CRM** (crm) – for cultural heritage objects, events, and provenance, especially for museum items, architectural monuments, and artworks.
- **BIBFRAME** (bf) – for bibliographic resources, including the manuscript *Codex Florentinus* and the modern edition of Procopius' *History of the Wars*.
- **Nomisma Ontology** (nmo) – for the numismatic item (gold solidus), covering denomination, mint, and issuing authority.
- **Schema.org** (schema) – for contemporary media productions (board game, symphony, film, video game) and general web resources.
- **FOAF** (foaf) – for describing people and their relationships.
- **GeoNames** (gn) – for geographical places.
- **Wikidata / VIAF** – used for linking to external authority records via owl:sameAs statements.

A full list of **namespace prefixes** adopted in the project is provided below:

- @prefix sits: <https://iustinianus-lod.github.io/Justinian-LOD//>
- @prefix aat: <http://vocab.getty.edu/aat/>
- @prefix bf: <http://id.loc.gov/ontologies/bibframe/>
- @prefix bibo: <http://purl.org/ontology/bibo/>
- @prefix crm: <http://www.cidoc-crm.org/cidoc-crm/>
- @prefix dbo: <http://dbpedia.org/ontology/>

- @prefix dcterms: <http://purl.org/dc/terms/>
- @prefix nm: <http://nomisma.org/id/>
- @prefix nmo: <http://nomisma.org/ontology#>
- @prefix owl: <http://www.w3.org/2002/07/owl#>
- @prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
- @prefix schema: <https://schema.org/>
- @prefix skos: <http://www.w3.org/2004/02/skos/core#>
- @prefix xsd: <http://www.w3.org/2001/XMLSchema#>
- @prefix wd: <https://www.wikidata.org/wiki/>

This combination of vocabularies ensures **flexibility** and **semantic precision**, allowing the dataset to be integrated with other Linked Open Data resources while remaining faithful to the original metadata structures.

KNOWLEDGE REPRESENTATION

CSV FILES IN NATURAL LANGUAGE

As a first step in the data creation process, we produced a collection of **.csv files**, with **one table for each of the ten selected items**, containing a complete description of the cultural heritage resources in **natural language**. Each CSV file follows a simple structure organized in **subject-predicate-object statements**, allowing the information extracted from the original metadata to be represented in a clear and consistent form before being converted into formal semantic data.

The goal of this phase was to translate the descriptive information provided by the institutions holding the items into a set of readable statements that highlight both the characteristics of each item and its relationships with other entities relevant to the project, such as

historical figures, places, events, or cultural contexts related to the reign of Emperor Justinian I.
For some items, as discussed in the **Metadata Analysis** section, the predicates used in the CSV files directly reflect the descriptive fields adopted by the institutions that provided the original metadata. In these cases, the natural language predicates correspond closely to the categories used by the data providers, ensuring that the original meaning and structure of the information are preserved.
For other items, however, the institutions did not provide a clearly defined metadata structure or used descriptive fields that were not immediately compatible with the modeling approach adopted in this project. In such cases, the mapping of predicates was carried out by the project team. These choices were guided by two main criteria: first, the **semantic accuracy of the predicate in relation to the type of entity being described**, and second, the **generalization of the relationship expressed between subject and object**, in order to facilitate the subsequent translation of the statements into formal language using established ontologies and vocabularies.
This intermediate natural language representation therefore plays an essential role in the workflow of the project. It acts as a bridge between the heterogeneous metadata provided by cultural heritage institutions and the formal semantic representation that will later be expressed through RDF triples and Linked Open Data technologies.

RDF PRODUCTION

From these natural language CSV files, another set of CSV files is created using formal language. These files will be found under the folder "csv_files/Formal_language" within the project's repository. The files will have the same structure as the natural language files, following the subject-predicate-object form, so that every statement could be traced back to its equivalent form in natural language and mapped into a semantic triple.

The formalization process consisted of two parallel processes. For the predicates, every identified relationship from the description of

the natural language was associated with a property from a predefined ontology or vocabulary, such as those used in conceptual modeling. For instance, Dublin Core properties (dcterms) were used for generic descriptive metadata, such as title, date, language, and source. BIBFRAME properties were used for bibliographic resources, CIDOC CRM for cultural heritage objects and events, Schema.org for media items and generic descriptions, and Nomisma Ontology for the coin. For the objects, every object is associated with a URI from authority control, rather than being represented as a string. For instance, VIAF is used for people and authors, Wikidata for historical concepts and entities, GeoNames for geographical locations, and Getty AAT for object types and materials.

Entities that are key to the project but not owned by a particular institution, like Justinian I, Theodora, Belisarius, the Byzantine Empire, and Corpus Iuris Civilis, have been given custom URIs under the base namespace of the project (sits:).

Two additional CSV files have been created to list relationships between entities that are not items themselves but are relevant to the items. These relationships are the relationships between historical figures and the items they are depicted on, the descriptions of the museums that the items are owned by, and the relationships between the Byzantine Empire and its capital city of Constantinople. In addition to these relationships, owl:sameAs statements have been added to ensure the custom URIs created by the project match the equivalent Wikidata entries.

Python script for transformation

The RDF dataset was created programmatically by executing a Python script that utilized the rdflib, csv, and glob libraries. It starts by initializing an RDF graph and specifying a dictionary of namespace prefixes and their respective URIs, which includes all the ontologies and vocabularies utilized throughout the project.

The script then reads the csv_files/Formal_language directory, parses all files with the .csv extension, and processes each CSV file by utilizing the DictReader, which reads the header row automatically. It then reads each subject-predicate-object statement in the CSV file. For each statement, the subject and predicate are expanded from CURIE notation to a complete URI by utilizing the prefix dictionary. Complete http and https URIs are processed directly, and prefix casing is standardized so that the same URI can be correctly resolved regardless of its representation in the CSV file.

Object types are determined based on their contents. Angle bracket URIs will be used as URI references, CURIEs will be expanded based on the prefix map, language-tagged literals will be converted into proper RDF language-tagged strings, four-digit values within a valid range of historical years will be mapped into xsd:gYear, numerical values will be mapped into xsd:integer, and all other values will be stored as plain string literals. If a subject or predicate cannot be resolved, it will be skipped and will appear in console output, making it quite easy to spot and fix errors in the original CSV files without interrupting the conversion process.

Once all files have been processed, the finished graph will be serialised in Turtle (.ttl) format into a file called full_dataset.ttl. Turtle was selected as the output format for its readability and its popularity among Linked Data tools and validators.

TEXT ANALYSIS

XML/TEI DOCUMENT

The encoding work focused on selected excerpts from **Procopius of Caesarea's *History of the Wars***, one of the most important historical sources for the reign of Emperor Justinian I. The encoded text is based on the English translation edited by **H. B. Dewing** for the *Loeb Classical Library*, as made available through the Internet Archive. Rather than encoding the entire work, the project focuses on

carefully selected passages that are particularly relevant to the political and military history of Justinian's reign.
In particular, the selected excerpts highlight episodes connected to Justinian's imperial policies, military campaigns, and the activities of key historical figures such as the general **Belisarius**. This approach allows the text to serve both as a readable digital edition and as a dataset suitable for **prosopographical and event-based analysis** within the broader Linked Open Data framework of the project.
The TEI encoding preserves structural and semantic features of the original source while enabling the identification of entities such as people, places, and events mentioned in the narrative. Page breaks from the original printed edition were maintained through the `<pb/>` element in order to preserve the layout and reference structure of the source text.
To ensure the correctness of the markup, the XML/TEI document was validated using online XML validation tools and the XML/TEI plugin for Visual Studio Code.
The XML/TEI document is structured into two main components: the **`<teiHeader>`**, containing metadata about the encoded text, and the **`<text>`** element, which contains the encoded content itself.

`<teiHeader>`

The `<teiHeader>` contains the metadata describing both the original source and the digital edition created for the project.

- **`<fileDesc>`** provides bibliographic information about the work, including the title, the author (Procopius of Caesarea), the translator and editor of the Loeb edition, the publisher, place of publication, date, and identifiers related to the digital copy hosted on the Internet Archive.
- **`<encodingDesc>`** describes the objectives of the digital encoding and explains the editorial choices adopted in the project. In particular, the selected passages were chosen to document the reign of Justinian and the activities of the individuals involved in the historical narrative. The section also clarifies that the encoding supports both human interpretation and the

extraction of structured information for Linked Open Data analysis.

- **<profileDesc>** contains additional contextual metadata. This section includes information about the languages used in the text, as well as structured lists of entities referenced in the encoded passages. In particular, the <listPerson> element records historical figures mentioned in the text, such as Justinian, Belisarius, and Theodora, while <listPlace> includes geographical locations related to the events described. These entities are assigned unique identifiers (xml:id) and, when possible, linked to external authority resources such as **Wikidata**, enabling semantic interoperability with external datasets.

Additionally, a set of **keywords** describing the thematic focus of the text, such as Byzantine history and military history, is included to facilitate thematic classification and retrieval.

<text>

The <text> element contains the encoded content of the work and is divided into structural sections.

- **<front>** includes the preliminary material of the edition, such as the title page and the introductory section describing the historical context of Procopius' work and his relationship to the court of Justinian.
- **<body>** contains the selected chapters and passages from *History of the Wars*. Within the encoded text, personal names are marked with <persName> elements, places with <placeName>, and temporal references with <date>. These elements are connected through references to the entities defined in the TEI header, ensuring consistent identification across the document and facilitating later semantic processing.

FROM XML TO HTML

To make the encoded text accessible in a readable format, an XSL stylesheet was created to transform the XML/TEI document into an HTML page. The stylesheet defines transformation rules that

identify elements in the XML source using **XPath expressions** and convert them into corresponding HTML structures.
The transformation process begins at the root element of the TEI document and recursively processes the hierarchical structure of the XML tree. Templates defined in the XSL file specify how particular TEI elements, such as titles, metadata sections, lists of people and places, and paragraphs of text, are rendered in HTML.
During this transformation, the metadata contained in the TEI header are reorganized and displayed in structured HTML sections. These sections present information about the original edition, the digital edition produced for the project, and contextual lists of people, places, and keywords referenced in the encoded text.
Specific templates were also implemented to process the textual content of the document, converting structural elements such as chapters and paragraphs into their HTML equivalents while preserving the semantic distinctions between different types of entities (e.g., people, places, and terms).
The approach adopted ensures a **clear separation between content and presentation**: the XSL stylesheet manages the transformation of the semantic structure of the TEI document, while visual formatting is handled through CSS rules embedded in the resulting HTML page. The final output is a well-structured and readable **digital edition of the selected passages from Procopius' work**, accessible through a standard web interface.
Once the XSL transformation rules were completed, both the **XML and XSL files were uploaded to the online tool** Free Formatter, which was used to perform the transformation and generate the corresponding HTML output. The resulting HTML file was then reviewed and tested to ensure that the encoded information was correctly displayed and navigable as a web page.

XML/TEI TO RDF

In order to extend the Linked Open Data model of the project to include the encoded textual source, a Python script was created to

convert the XML/TEI document of the Procopius encoding into an RDF dataset.

In addition to creating a Linked Open Data model of the encoded source, the process of creating an RDF dataset was also designed to leverage the structured metadata that was encoded in the TEI document. For example, entities mentioned in the teiHeader of the TEI document, such as historical figures mentioned in the listPerson element and locations mentioned in the listPlace element, were extracted and assigned a URI in the project's base namespace. If the TEI document encoded references to external authority files, such as Wikidata or VIAF, these were also reused directly as owl:sameAs statements in the resulting RDF dataset.

The descriptive properties were represented using the same ontologies used throughout the entire project, which were **dcterms** for bibliographic details and **foaf** for describing individuals. The relationships between entities found in the text, e.g., the association between Belisarius and Justinian, or the association between the historical characters and the geographical places mentioned in the text, were represented as RDF triples, expanding the graph with new information directly extracted from the encoded source.

The resulting graph was then serialized to produce an RDF file in Turtle format, which complements the original dataset and can be queried and browsed alongside it. Validation was performed using the **W3C RDF Validator tool** to ensure the syntactic correctness of the produced file before publication.